



Optimizing Yelp's Ad Relevance: Insights from the Box Search Experiment

BAMS 521 Group 21:

Yousef Jafarnia Dabanloo 37746765

Ricky Liang 1043691

Connor Lyons 28416303

Ronghao Wu 19947225

Sisly Lyu 72710643



Introduction & Experiment Overview

Business Background

- Yelp derives 96% of its revenue from ads.
- Users performing map searches have shown lower engagement—likely because ads are not geo-targeted.

Experiment Purpose

- **Hypothesis:** Enforcing geographical constraints (showing only ads within the map-defined area) increases ad relevance and user engagement.
- **Experiment:** Treatment group sees only ads within map constraints; control group sees all relevant ads
- **Trade-Off:** Stricter geo-constraints may reduce the available ad inventory and overall revenue.

Metrics Monitored

- **Primary Metric:** Clickthrough Rate (CTR)
- **Secondary Metrics:** Advertising Shown Rate, Average Revenue per Click, Average Revenue per Eligible Search



Exploratory Data Analysis

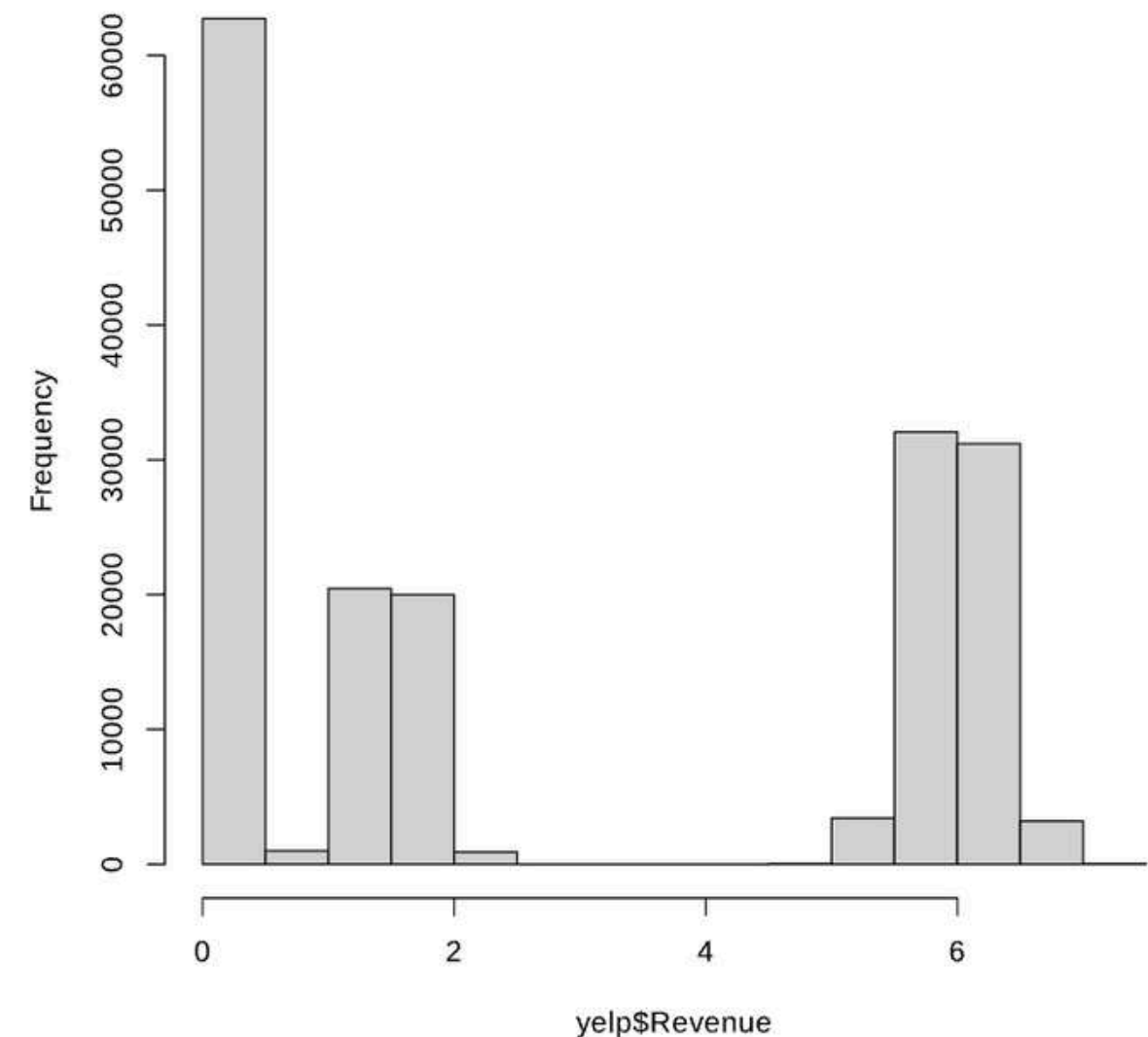
Yelp Data Distributions



Key Insights from Yelp Data

- 60% of searches are for local services – high value segment
- 70% of users search on mobile devices – UX-critical platform
- 95% of searches are eligible for ads – large monetization base
- 80% of eligible ads are shown – strong ad exposure
- 64% of shown ads are clicked – high engagement rate

Histogram of yelp\$Revenue



Revenue Distribution Observations

- **Bimodal distribution:** Peaks at \$0 and \$6
- **\$0 peak:** Likely from non-clicked ads
- **\$6 peak:** Suggests strong value from local service ads
- Smaller cluster near \$1.50 may reflect mid-tier ad types
- Confirms multiple ad value tiers and importance of local relevance



Results and Interpretation — Conversion Rate Analysis

Control

94.99%



94.25%



75.92%



67.98%

Treatment

95.17%



73.96%



85.74%



60.34%

Eligible -----

Shown -----

Clicked -----

Overall conversion rate is significantly lower in the **treatment group (60.34%)** compared to the **control group (67.98%)**, $p < 0.001$.

Breaking down the conversion funnel:

1. **Eligible Rate(Eligible/Total)**: Treatment shows a **slight improvement**, is **not statistically significant** ($p = 0.090$).
2. **Ad Shown Rate(Shown/Eligible)**: Treatment **performs substantially worse**, **highly statistically significant** ($p < 0.001$).
3. **CTR(Clicked/Shown)**: Treatment demonstrates **higher click-through rate**, which is **statistically significant** ($p < 0.001$).

In summary, **Ad shown Rate** is the primary bottleneck in the treatment group's performance. **CTR** in treatment group's improvement isn't enough to overcome the deficit at the ad shown stage.

Results and Interpretation — Revenue Impact Analysis

Control

\$3.00

\$3.16

\$3.35

\$4.41

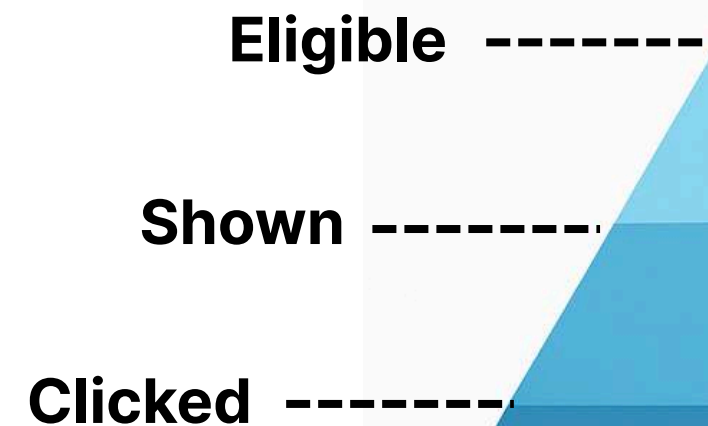
Treatment

\$2.52

\$2.65

\$3.58

\$4.18



For each yelp search, control: \$3.00; Treatment: \$ 2.52, breaking down:

Revenue per Eligible Search:

- **Control:** \$3.16; **Treatment:** \$2.65 (16% lower)
- The treatment group generates significantly less revenue per eligible search.
- **Cause:** Low ad shown rate creates a major bottleneck in the funnel.

Revenue per Shown Ad:

- **Control:** \$3.35; **Treatment:** \$3.58 (7% higher)
- When ads are actually shown, the treatment group outperforms the control group in revenue generation.
- **Cause:** Higher treatment value may suggests greater loss during conversion.

Revenue per Clicked Ad:

- **Control:** \$4.41; **Treatment:** \$4.18 (5% lower)
- Once clicked, control group ads generate slightly more revenue per click.
- **Cause:** Lower conversion value despite higher CTR suggests audience targeting issues.

Aggregate Results Overview

Aggregate Results by Treatment

Metric	Control	Treatment	Change	p-value
Clickthrough Rate (CTR)	75.9%	85.7%	+9.8 pp	< 0.0001
Ad Shown Rate	94.3%	74.0%	−20.3 pp	< 0.0001
Revenue per Click (USD)	\$4.41	\$4.18	−5.2%	< 0.0001
Revenue per Eligible Search (USD)	\$3.16	\$2.65	−16.1%	< 0.0001

Although the treatment improves CTR considerably, it comes with a substantial reduction in ad impressions and overall revenue.



Engagement Analysis: Local vs. Non-Local Services

Results by LocalService

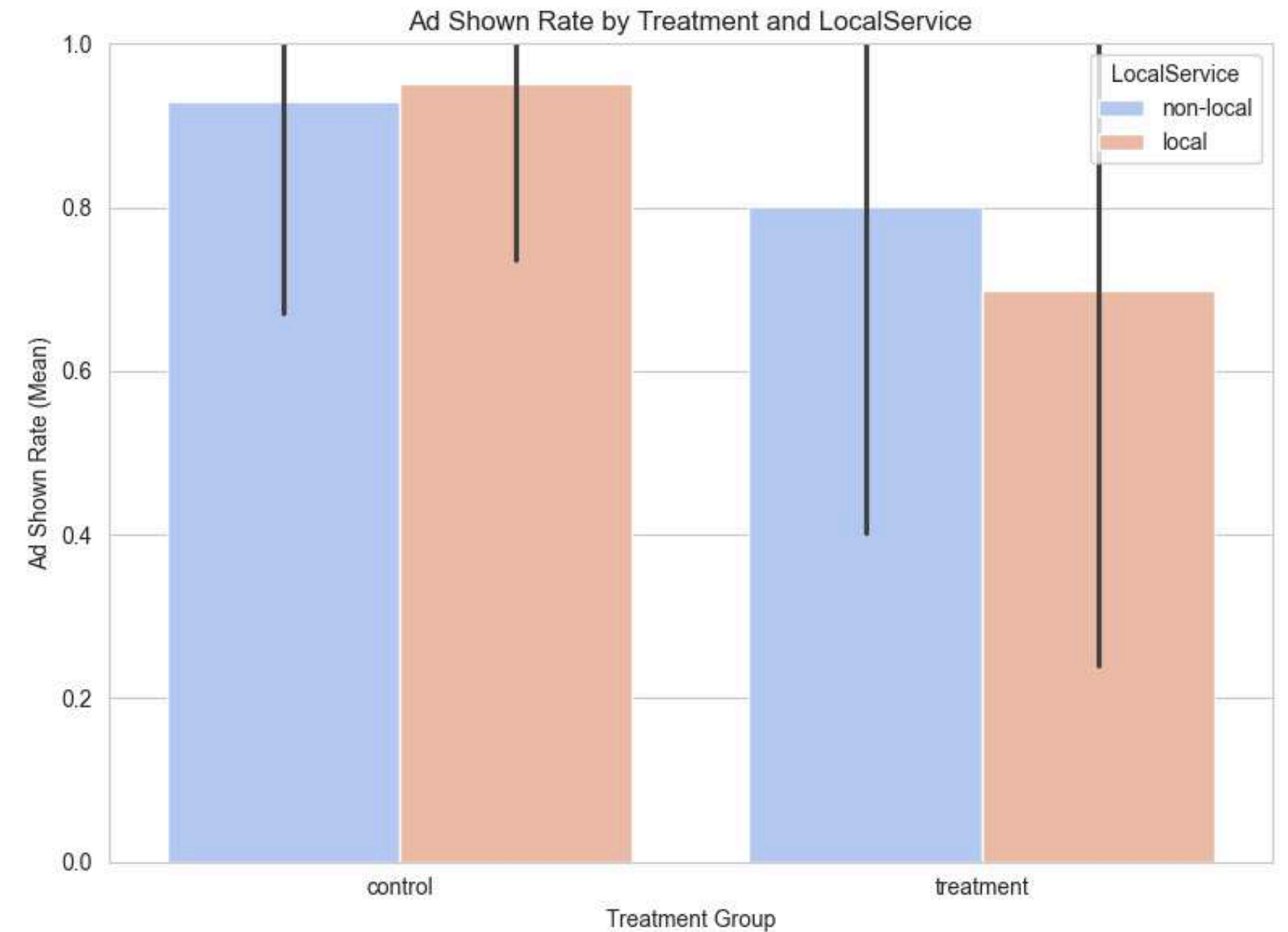
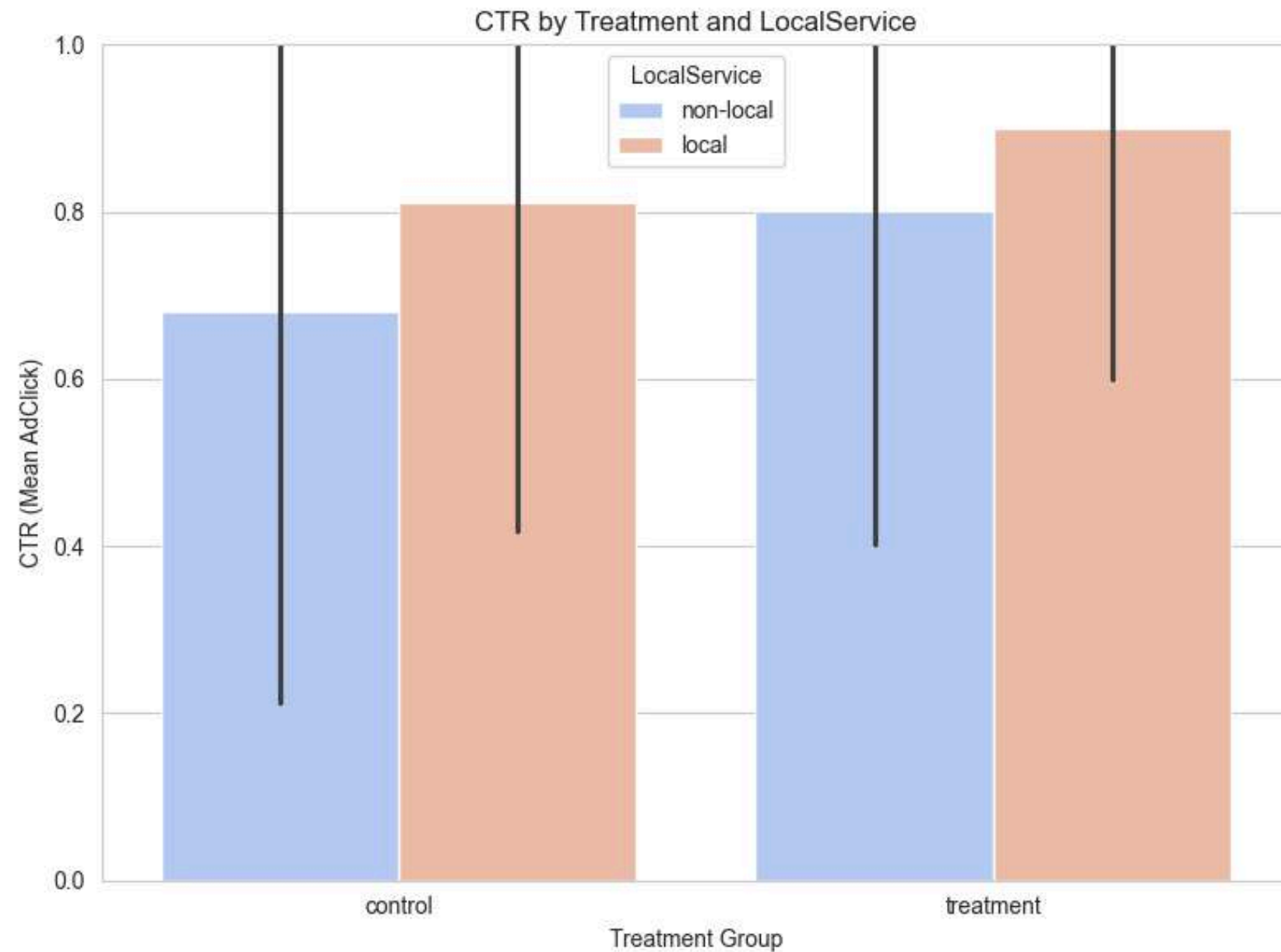
Metric	Local (Control)	Local (Treatment)	Non-Local (Control)	Non-Local (Treatment)
CTR	81.1%	90.0%	68.0%	80.2%
Ad Shown Rate	95.2%	69.9%	92.9%	80.1%

- Local searches have a higher baseline CTR.
- Under treatment, the ad shown rate for local searches drops by 25.3 percentage points versus 12.8 pp for non-local searches.

The loss of ad inventory in high-value local searches is likely a primary driver of the revenue decline.



Engagement Analysis: Local vs. Non-Local Services



Revenue Impact Analysis

Revenue Breakdown by LocalService

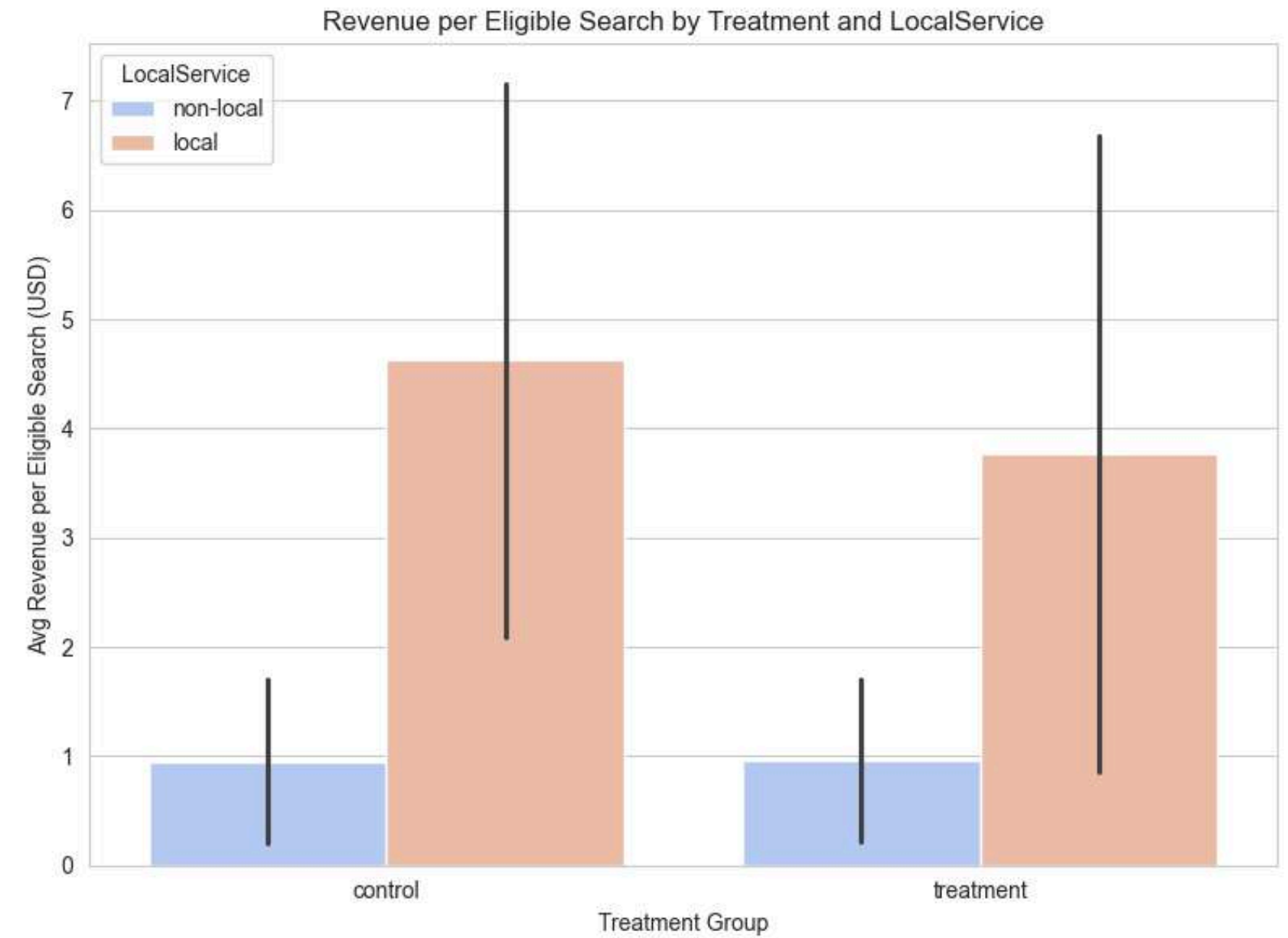
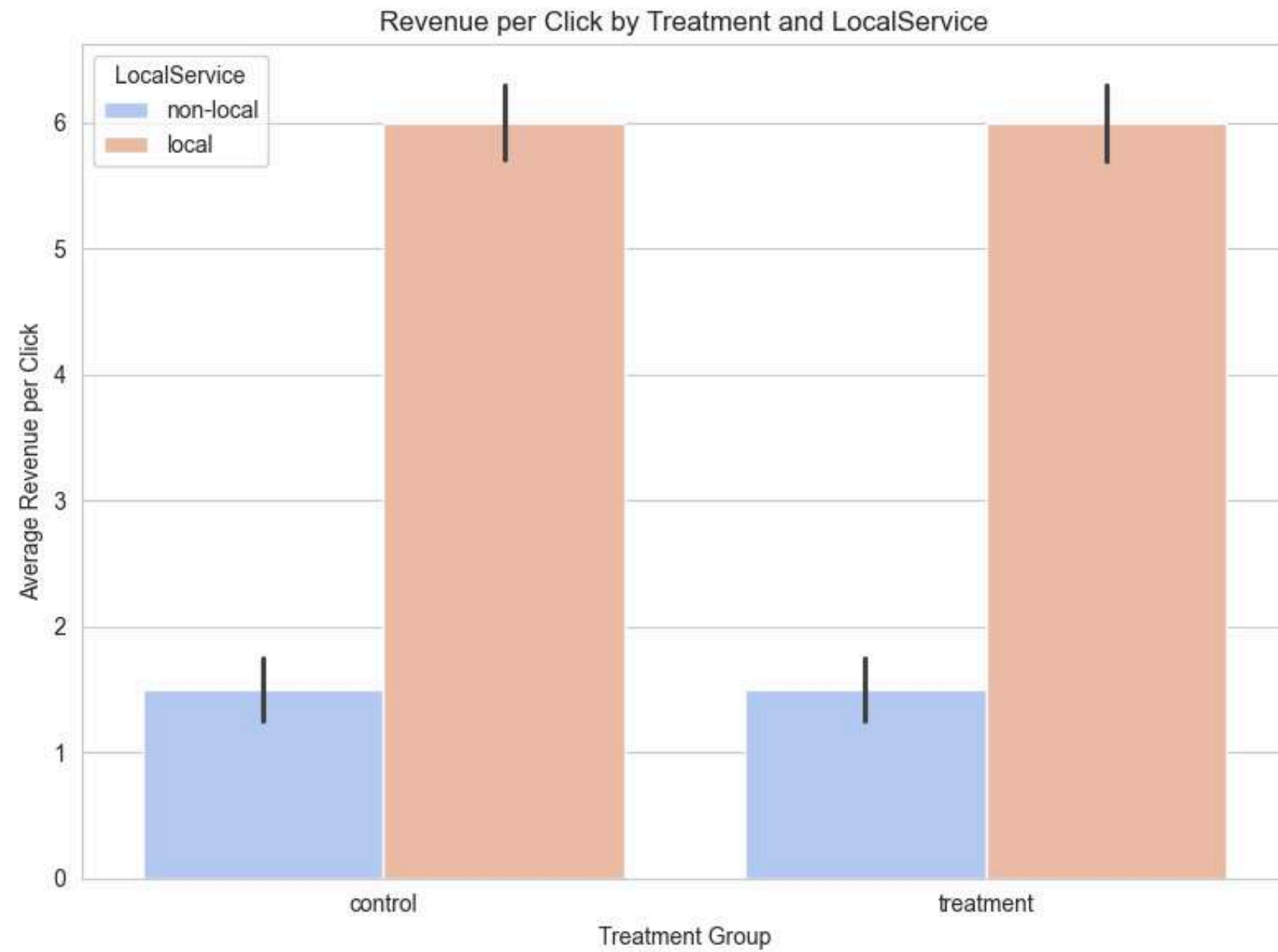
Metric	Local (Control)	Local (Treatment)	Non-Local (Control)	Non-Local (Treatment)
Revenue per Click (USD)	\$6.00	\$6.00	\$1.50	\$1.50
Revenue per Eligible Search (USD)	\$4.63	\$3.77	\$0.95	\$0.96

- Revenue per click remains nearly unchanged across treatment for both local and non-local searches.
- However, overall revenue per eligible search drops significantly for local searches (−18.6%) under treatment.

Reduced ad exposure in local searches—despite high engagement on those clicks—is the major factor in the revenue loss.

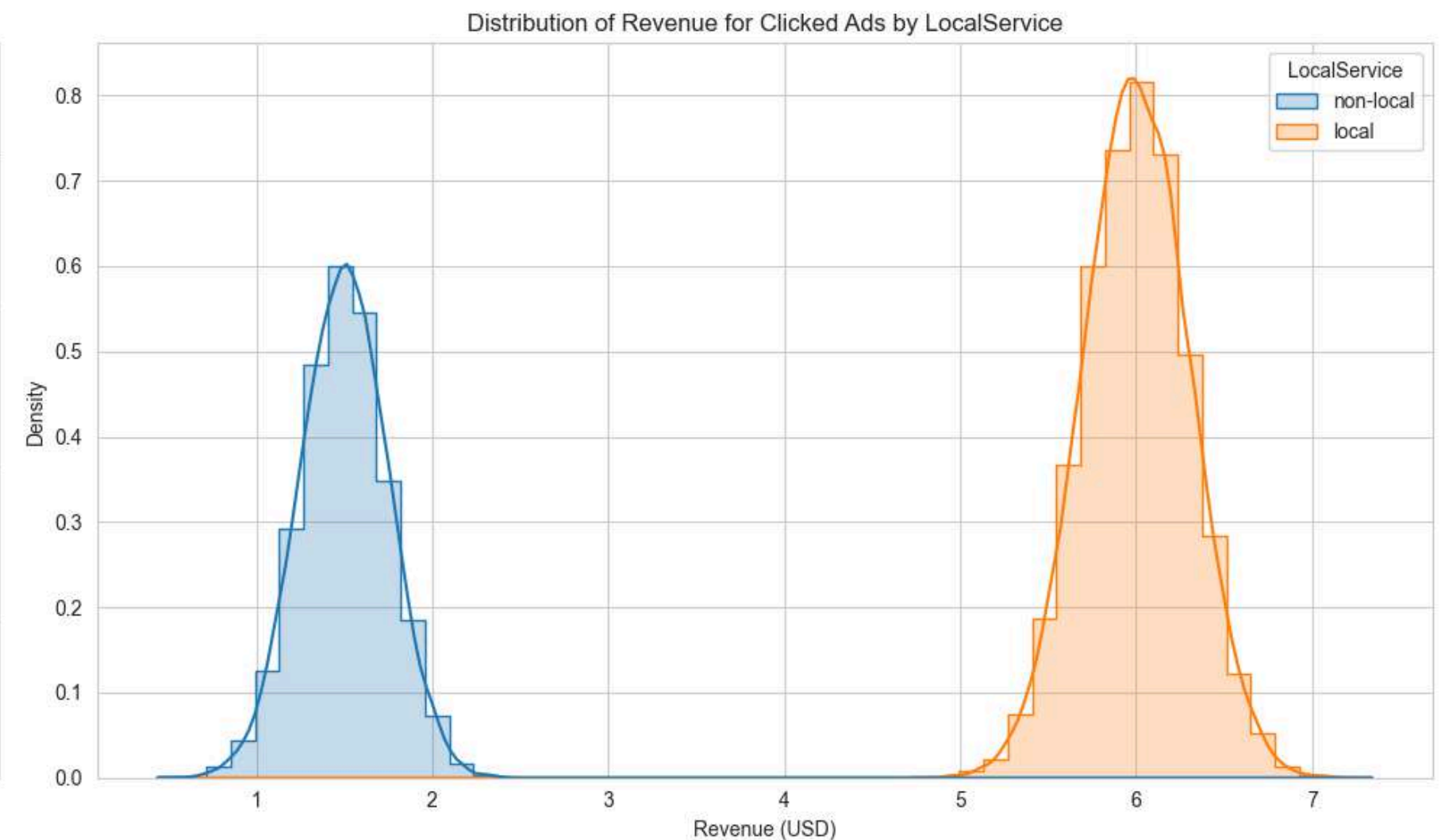
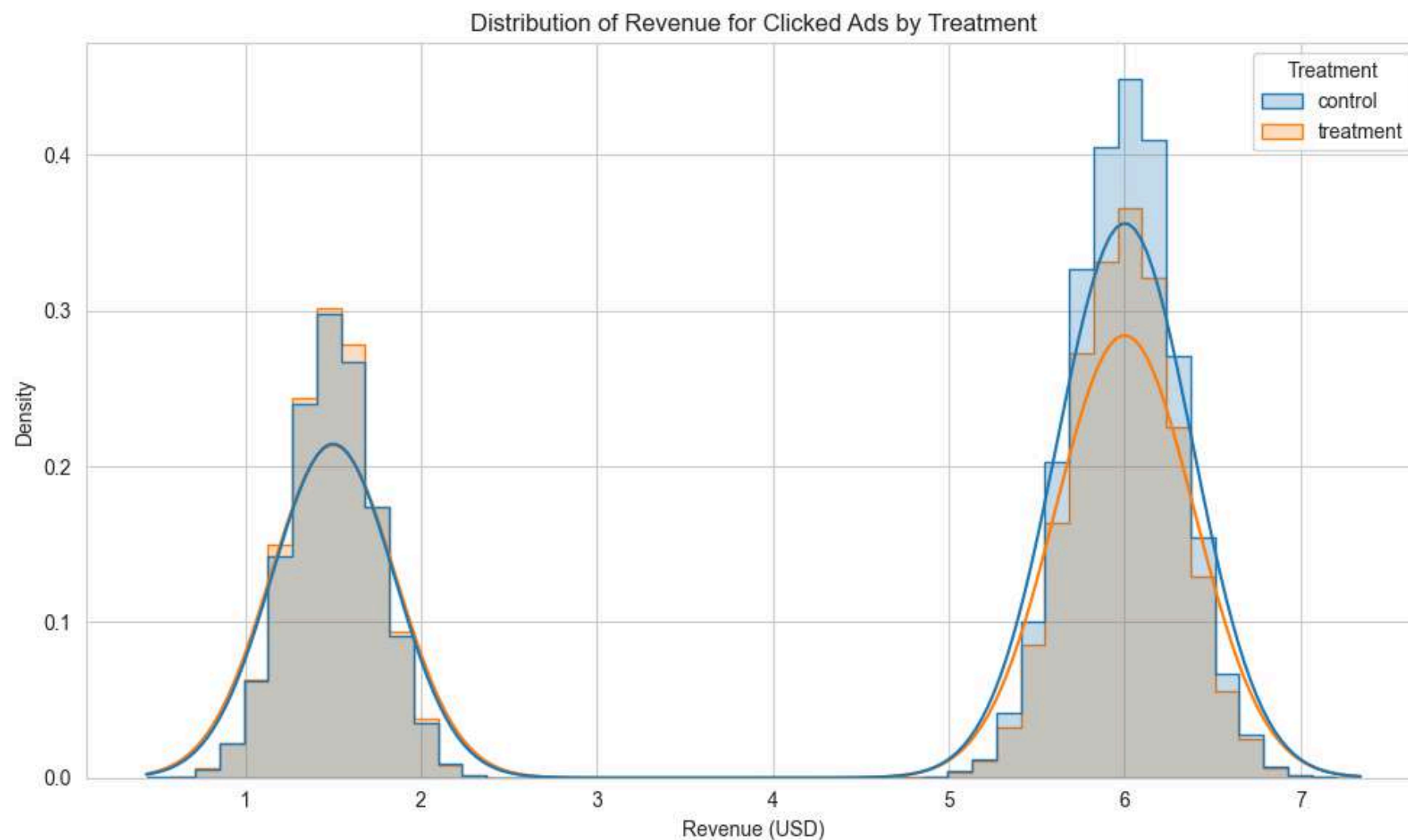


Revenue Impact Analysis



Revenue Impact Analysis

We can see that the high revenue ads are for local services and the decrease in the treatment revenue is mainly due to high revenue ads.



Recommendation

Recommendation 1: Hybrid Geo-Constraint Approach

Action:

- Enforce strict geo-constraints only when ad inventory is sufficient.
- Dynamically relax geo-constraints when ad inventory is low or auction results would yield low revenue.

Benefit:

- Restore ad exposure in cases where it is critically low, balancing quality with quantity.

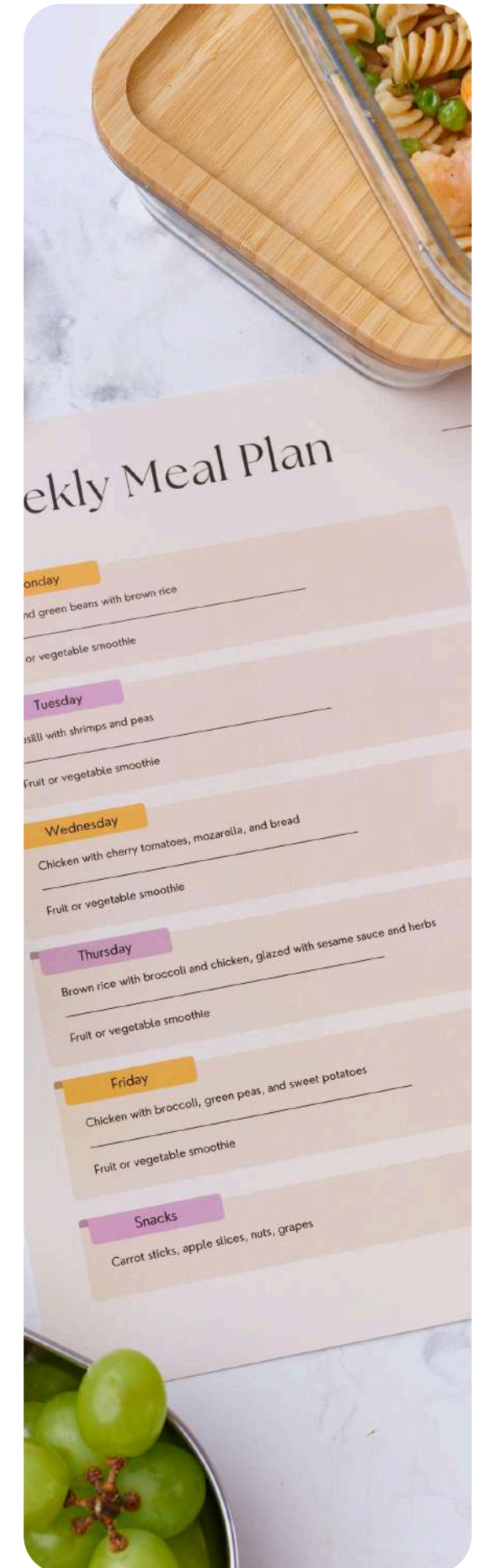
Recommendation 2: Differential Strategy for Local Searches

Action:

- Apply softer geo-constraints for local service searches where revenue per click is high.

Benefit:

- Protect high-value inventory and maintain revenue despite the push for better relevance.



Recommendation

Recommendation 3: Dynamic Auction & Inventory Management

Action:

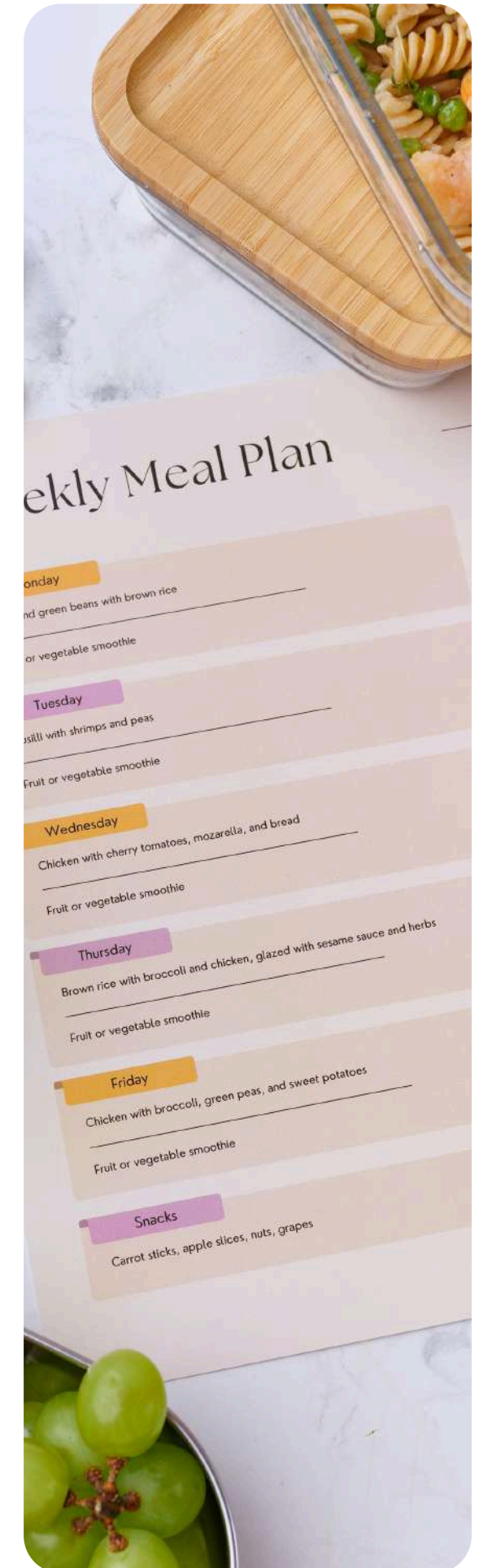
- Adjust auction parameters (e.g., reserve prices) and inventory rules based on real-time data.

Benefit:

- Optimize revenue outcomes while preserving user engagement.

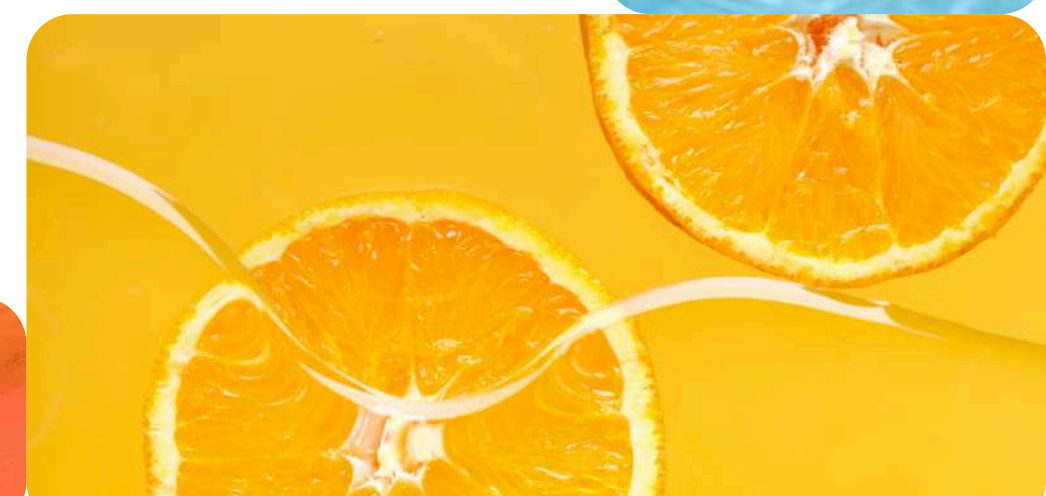
Next Steps:

- Pilot the recommended hybrid and differential strategies with A/B testing.
- Monitor the key metrics and iterate quickly in a build–measure–learn cycle.
- Validate all recommendations with additional experiments before full-scale rollout.





Thank You!



Appendix Slide A: Statistical Outputs

Two-way ANOVA Results for CTR, Ad Shown Rate, Revenue per Click, Revenue per Eligible Search; moderator analysis

Metric	Factor	Sum Sq	F-Statistic	p-value
CTR	Treatment	363.074	2376.006	< 0.0001
	LocalService	455.021	2977.722	< 0.0001
	Treatment x LocalService	8.746	57.236	3.89×10 ⁻¹⁴
Ad Shown Rate	Treatment	1713.873	14019.361	< 0.0001
	LocalService	63.519	519.580	< 0.0001
	Treatment x LocalService	155.321	1270.517	< 0.0001
Revenue per Click	Treatment	0.006	0.079	0.7787
	LocalService	532623.071	6703800.000	< 0.0001
	Treatment x LocalService	0.002	0.025	0.8744
Revenue per Eligible Search	Treatment	10701.875	2280.442	< 0.0001
	LocalService	420112.661	89521.000	< 0.0001
	Treatment x LocalService	7561.230	1611.208	< 0.0001